

ABSTRACT: Grip strength reference values for a portable dynamometer, the hand-held Vigorimeter, were calculated, and its validity, reliability, and responsiveness were examined in patients with immune-mediated polyneuropathies. We studied 530 healthy controls (age 5–93 years), 113 patients with stable polyneuropathy (83 with Guillain-Barré syndrome [GBS], 22 with chronic inflammatory demyelinating polyneuropathy [CIDP], and 8 with a gammopathy-associated polyneuropathy), and 20 patients with GBS or CIDP and changing clinical conditions (longitudinal group). An arm-disability scale was also assessed. Grip-strength reference values were calculated depending primarily on age and gender. Significant association was obtained between the Vigorimeter and the arm scale (in stable group, Spearman rank: $r = -0.52$ to -0.62 ; $P \leq 0.0005$; in longitudinal group, linear regressions: $r = 0.62$ – 0.64 , $P < 0.0001$). Good interobserver and intra-observer agreements (analysis of variance, $r = 0.95$ – 0.97) and high responsiveness (standardized response mean scores ≥ 0.8) were demonstrated for the Vigorimeter. These results emphasize the clinical usefulness of the Vigorimeter, particularly in patients with immune-mediated polyneuropathies.

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ASSESSING GRIP STRENGTH IN HEALTHY INDIVIDUALS AND PATIENTS WITH IMMUNE-MEDIATED POLYNEUROPATHIES

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Accepted 10 April 2000

The Medical Research Council (MRC) scale is primarily used to assess muscle strength in neuromuscular disease.²⁶ Interpreting assessments by the MRC scale is sometimes complex, because this scale has a nonlinear pattern with a broad range. Perhaps the biggest disadvantage of the MRC scale is that not all muscles can be appropriately measured with it. An example is assessing strength of the small muscles

embracing hand function. This is particularly of importance when distal weakness predominates, as generally occurs in patients with polyneuropathies such as Guillain-Barré syndrome (GBS) and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP). Especially when evaluating therapeutic responses, assessment of muscles involving hand function gives important information. Grip strength, reflecting distal strength and upper limb function, is a prognostic indicator of clinical and functional recovery and is useful in monitoring the effect of treatment.^{21,28,29,33,35}

Among the instruments developed to measure grip strength (GS) quantitatively, portable dynamometers are the most popular among neurologists and rehabilitation physicians. One of these instruments is the Vigorimeter (Martin, Tuttlingen, Germany).^{6,15} Reference values for this instrument have

*Listed at the end of this article.

Abbreviations: ADS, arm-disability scale; CIDP, chronic inflammatory demyelinating polyradiculoneuropathy; GBS, Guillain-Barré syndrome; GS, grip strength; IVIg, intravenous immunoglobulin; kPa, kiloPascal; MGUSP, polyneuropathy associated with a monoclonal gammopathy of undetermined significance; MRC, Medical Research Council; SRM, standardized response mean

Key words: grip strength; immune-mediated polyneuropathies; reference values; reliability; responsiveness; validity

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been provided for right-handed healthy individuals.^{8,9,12,14,24,30,34} However, in these studies, various age groups were investigated with varying methodologies. For example, anthropometric data were not systematically examined, number of trials for each hand and resting periods between trials varied considerably, and various arm/hand/body positions were used. These observations hamper the general applicability of the obtained normal values. To date, no study has covered the evaluation of GS by the Vigorimeter in ages ranging from childhood to the elderly, and only a limited number of papers have addressed the clinical usefulness of this approach.^{5,20,21,28,35} The Vigorimeter has been applied only twice in patients with immune-mediated polyneuropathies.^{21,35} A good correlation was obtained between GS and MRC sum-score values in a group of 11 patients with GBS, thus demonstrating the validity of the Vigorimeter.²¹ However, a more extensive clinical evaluation of the Vigorimeter is required in patients with immune-mediated polyneuropathies before its general use as an impairment outcome measure in these disorders is recommended.^{11,19,36}

Prompted by these considerations, we collected GS values and personal variables (length, weight, hand circumference, and hand dominance) in a large group of healthy controls with a wide age span to determine new reference GS values. Additionally, the validity, reliability, and responsiveness of the Vigorimeter were investigated in patients with GBS, CIDP, or a polyneuropathy associated with a monoclonal gammopathy of undetermined significance (MGUSP).^{11,13,19,22} The obtained GS values in the patients group were related to an arm-disability scale (ADS).³¹ This was essential not only for the validation of the Vigorimeter but also to investigate whether GS could be used as an indicator of disease activity at the disability-outcome level.³⁶

PARTICIPANTS AND METHODS

Healthy Controls. We recruited 530 healthy controls from hospital personnel, relatives, and friends accompanying patients at our outpatient clinic, and healthy elderly, living in the village of Nieuw Vennep as part of the municipality of Haarlemmermeer, The Netherlands. There was a door-to-door mailing explaining the purpose and significance of the study. Attempts were made to recruit participants representing a wide variety of social and occupational backgrounds. Children and adolescents were recruited at one primary and one high school. The eligibility criteria were: normal cognitive function, independence in activities of daily living, socially active, preserved vision, absence of any impairment af-

flecting upper limb function, and normal physical development in children as reported by parents. Neurological examination was performed in all participants, with special regard to upper limb functions. Healthy individuals were stratified for age and gender, and 10 age groups (5–9, 10–14, 15–19, 20–29, . . . , 70–79, and ≥ 80 years) were formed, each group consisting of 50–55 participants. The first three age groups had a smaller age range, in order to more accurately register the rapid change in GS during growth.

Patients. We recruited 113 clinically stable patients (83 GBS, 22 CIDP, 8 MGUSP) from the Rotterdam immune-mediated polyneuropathy databank and the Dutch GBS study group (stable group). These selected patients still had residual signs or symptoms of their illness, representing a broad range of disability. Nine CIDP patients required interval treatment with intravenous immunoglobulins (IVIg) ranging from weeks to months. With this therapy, the patients' clinical condition was stable for more than 6 months.

Twenty consecutive patients with recently diagnosed GBS ($n = 7$) or CIDP ($n = 13$) were enrolled to investigate the responsiveness of the Vigorimeter to clinical changes over time (longitudinal group). All patients with GBS and CIDP met the international research criteria for their illness.^{1,4} The diagnosis MGUSP was established after excluding all possible underlying causes for the gammopathy and polyneuropathy.²⁷ Patients were excluded from participation if there was any concomitant disease that might interfere with general strength and physical functioning.

Assessment Tools/Scales. The Vigorimeter is an instrument that has been used to measure GS, although, strictly speaking, it measures the air pressure in the bulb and not force (Fig. 1).^{6,8,9,12,14,15,20,24,30} The pressure in the bulb is registered on a manometer via a rubber junction tube and expressed in kiloPascals (kPa). According to the manufacturer's recommendation, the small bulb was used in the ages up to 10 years and the medium-sized bulb, in the remaining participants.

The ADS scale comprises a good functional description of the arms in a checklist form suitable for interviewing patients.³¹ Daily arm activities such as dressing the upper part of the body, doing/undoing buttons and zippers, washing and brushing hair, using a knife and fork, and turning a key in a lock are scored as being "not affected," "affected but not prevented," or "prevented." Subsequently, these results



FIGURE 1. Photograph showing the Vigorimeter with the medium-sized rubber bulb placed comfortably in the right hand of a normal volunteer.

are translated into an arm grade. The score ranges from 0 (normal arm abilities) to 5 (severe symptoms and signs in both arms preventing all purposeful movements). Hence, this scale provides a general score for both arms. The ADS is a subset of the more comprehensive Guy's neurological disability scale.³¹

Test Procedures. *General Aspects.* All participants including the parents of the children gave informed consent for the study. All assessments took place in a quiet and temperature-controlled room (approximately 20°C) during daytime. Participants were examined according to the assessment recommendations of the American Society of Hand Therapists.³ All children were seated in appropriately sized chairs that allowed their feet to be flat on the floor. The investigator placed the child in the standardized position and encouraged the child to remain in that position. The bulb was positioned in the palm of the participant's hand with the air tube extending out

between the individual's thumb and index finger and with the fingers wrapped around the bulb so that the fingers touched the surface of the bulb as much as possible. Three GS measurements with maximum voluntary contractions for each hand were taken in alternating order. Between each trial, a pause of 30 s was assigned. The results of three trials for each hand were averaged and considered the GS score for that particular hand. Half the controls in each age group started with right-hand-grip assessment and the other half, with left-hand-grip measures, to control for the effect of order.

Before using the Vigorimeter, healthy controls were interviewed by one investigator (I.S.J.M.). Anthropometric data (body weight, height, and hand circumference) and hand dominance were collected. Hand circumference was assessed using a measuring tape that was circularly applied at the distal ends of the metacarpals of digits 2 to 5. For the purpose of this study, hand dominance was defined as the preferred hand for use. In children, hand dominance was determined by asking the participants to throw a ball or use a writing implement. Examination of the controls took place at our outpatient clinic or at the homes of the elderly; children were investigated at their schools.

Validity and Reliability. For the assessment of validity and reliability of the Vigorimeter in the stable group of 113 patients, 28 different pairs of examiners were formed by two neurologists and six residents in neurology. Prior to the study, all investigators received instructions in assessing the outcome measures. Twenty-seven ("variable") couples examined 68 patients (2 to 3 patients per couple). The remaining 45 patients were investigated by the "experienced" couple (I.S.J.M. and J.P.A.S.). These two authors investigated the effect of training and thus a possible increase in interobserver and intraobserver reliability in examiners who use the Vigorimeter more often.

The patients were examined on two different occasions at our outpatient clinic. During the first visit, the two members of an appointed pair performed their scores independently and consecutively (usually within 2 h) (interobserver measures). Within 2–4 weeks, the patient returned for a second visit, and one investigator of the earlier-assigned pair examined the patient again, without having access to previous results (intraobserver values). The assessments sequence at entry and the examination at the second visit were equally distributed among members of an assigned couple. Overall, each member of a couple examined the same number of patients as did his/

her partner. The ADS was measured only once by the first examiner in each patient.

Responsiveness. Twenty patients were longitudinally examined by the same clinician (I.S.J.M.), and the Vigorimeter and ADS were assessed at study entry and 8–13 times in each patient during follow-up. There was a standard follow-up schedule (weeks 0, 2, 4, 8, 12, 16, 21, 26, 32, 40, and 52), with additional clinical investigations if necessary. At each visit, the patients were requested to judge whether their clinical condition deteriorated (grade 1), remained stable (grade 2), or improved (grade 3) when compared with the last visit (“clinical-judgment scores”). The study took place between March 1997 and November 1998 and was part of a study on clinical outcome measures in patients with immune-mediated polyneuropathies on behalf of the Inflammatory Neuropathy Cause And Treatment (INCAT) group.

Statistics. Reference Values. Grip-strength reference values (median and 0.05 quantile values, corresponding to a specificity of 95%) were calculated in the healthy controls, depending primarily on age and sex, using quantile regression analyses with restricted cubic spline functions on age.^{16,18} These methods were performed to overcome the skewed distribution of GS values conditional on age and sex and to identify the regression curves that best fit the data.^{16,18} Quantile regression analyses were also performed in the control group to determine which personal variable(s), in addition to age and gender, best predict GS. In each gender, these regressions were performed separately for the dominant and nondominant hands. These GS values in each gender were the dependent variables, and age, hand circumference, length, and weight were the independent variables.

Validity and Reliability. In the stable group, correlation between GS and ADS was analyzed using Spearman rank correlation test. Also, random-effects linear regression analyses of GS on the ADS values were performed in the longitudinal group, taking into account the association of the data caused by the longitudinal structure. The latter was achieved using the program “xtreg” in STATA 6.0 (StataCorp. 1997, Stata Statistical Software: Release 5.0, College Station, Texas), which is based upon a cross-sectional time-series regression model as described by Dwyer and Feinleib.¹⁰ The interrater and intrarater reliability for the obtained GS values in the stable group of patients was quantified by estimation of the intraclass correlation coefficient using a one-way random-effects analysis-of-variance model for

the two investigator (“experienced” and “variable”) groups.

Responsiveness. Responsiveness was investigated by calculating the standardized response mean (SRM) scores for GS at various arbitrarily chosen occasions during follow-up (weeks 4, 12, 26, 40, and 52).²³ The SRM is equal to the mean change in score divided by the standard deviation of the change in score ($SRM = \mu_i - \mu_0 / SD(\mu_i - \mu_0)$; μ_i = mean GS value of the longitudinally examined group at week = i ; μ_0 = mean GS value at week = 0 [entry]).²³ A value between 0.5 and 0.8 is considered moderate and 0.8 or greater, high responsiveness.^{7,23} For each patient, the differences between all two-consecutive GS values were calculated (GS at visit i minus GS at visit $[i - 1]$). These differences were associated with the corresponding clinical-judgment scores (deterioration, stable, or improvement) using “xtreg.”¹⁰ All analyses were performed using Stata 6.0 for Windows 95. A value of $P \leq 0.05$ was considered statistically significant.

RESULTS

General Aspects. The whole procedure of interviewing, obtaining personal data, and assessing GS with the Vigorimeter took 5–10 min in each healthy participant. According to both investigators and participants, the Vigorimeter was easily administered and used.

Reference values for the Vigorimeter. Of the 551 healthy controls who were interviewed, 530 (275 females; 255 males) were enrolled in the study. Twenty-one individuals were excluded from participation, based on focal abnormalities (e.g., deformities of the forearm due to fracture, rheumatoid arthritis, Dupuytren’s contracture, hemiparesis, and signs of a polyneuropathy). The descriptive data of the healthy participants are presented in Table 1. Most individuals were right-handed (91%). There were almost no differences in GS values between the dominant and nondominant hands in the whole group of healthy controls (overall median differences: healthy males, 1 kPa; healthy females, 3 kPa) (Fig. 2). Based on these findings, only one graph was further developed for each gender separately to obtain reference values for both hands (Table 2).

Maximum median GS was observed in both genders at around 30 years. Males were consistently stronger than females (Table 2). Anthropometric variables (hand circumference, length, and weight) were consistently higher among men than women (Table 1). Quantile regression analyses showed that the median GS values for both sides were signifi-

Table 1. Descriptive data of healthy controls.

	Males			Females		
No. of participants	44	40	171	48	41	186
Age range (yr)	5–12	13–19	20–93	5–12	13–19	20–93
Mean (SD)	8.5 (2.4)	15.7 (1.9)	52.6 (19.9)	8.5 (2.4)	15.6 (1.7)	54.2 (20.4)
Hand dominance						
Right	40 (91%)	32 (80%)	156 (91%)	42 (88%)	35 (85%)	175 (94%)
Left	4 (9%)	8 (20%)	15 (9%)	6 (12%)	6 (15%)	11 (6%)
Length						
Mean (SD)	138.0 (15.7)	178.4 (10.4)	178.9 (8.3)	137.5 (15.7)	166.8 (9.3)	165.5 (7.8)
Range (cm)	108–168	155–195	159–203	111–169	131–181	146–186
Weight						
Mean (SD)	32.0 (10.5)	62.1 (11.6)	79.1 (9.8)	32.1 (11.7)	56.9 (10.0)	66.8 (10.2)
Range (kg)	17–60	37–85	55–108	16–62	32–75	46–104
Hand circumference						
Mean (SD)	17.1 (1.6)	21.1 (1.4)	22.5 (1.0)	16.3 (1.6)	18.8 (1.0)	19.7 (1.0)
Range (cm)	14–20.5	17.7–23.5	20.5–26.5	13.8–20.5	16–20.5	16.5–22.3

cantly associated with age in both genders and with hand circumference in men only ($P < 0.0001$).

Validity and Reliability of the Vigorimeter. At the beginning of the study, the stable group of patients (54 females; 59 males; median age, 56 years; range, 14–84 years) had a median duration of symptoms of 5.1 years. Of the patients, 96% were right-handed. The median right-hand GS value in this group was 67 kPa at first and third assessments (range: first assessment, 0–156; third assessment, 0–152) and 65 (range: 0–158) kPa at second assessment. The median left-hand GS value was: 64 (range: 0–158), 62 (range: 0–160), and 62 (range: 0–158) kPa at first, second, and third assessments, respectively. The median ADS was 2 at entry (range: 0–4). Seven patients were bed-bound and 14 required assistance or a device to walk over short distances. The remaining 92 patients were able to walk independently.

A moderate to good correlation was found between the Vigorimeter and the ADS values, thus demonstrating the validity of the Vigorimeter in this group of patients (experienced couple: GS right hands versus ADS: $r = -0.54$, $P = 0.0001$; left hands versus ADS: $r = -0.52$, $P = 0.0003$; variable couples: GS right hands versus ADS: $r = -0.57$, $P < 0.0001$; left hands versus ADS: $r = -0.62$, $P < 0.0001$). High interobserver and intraobserver agreements were also demonstrated for the Vigorimeter (experienced couple: interobserver GS right hands, $r = 0.97$; left hands, $r = 0.96$; intraobserver right hands, $r = 0.96$; left hands, $r = 0.95$; variable couples: interobserver and intraobserver values for both hand sides: $r = 0.97$, except for intraobserver left hands: $r = 0.96$; $P < 0.0001$ for all agreements).

Responsiveness of the Vigorimeter. Eight females and 12 males (median age, 54.0 years; range, 15–70

years) were examined longitudinally. All were right-handed. At study entry, four patients were bed-bound, one requiring artificial ventilation, and nine patients were unable to walk independently. All patients had experienced general loss of strength. During a follow-up period of 32 to 58 (median, 48.5) weeks, 186 visits were completed. Thirteen patients completed a 1-year follow-up. Except for one GBS patient, who experienced only mild symptoms, all patients received initial treatment with IVIg (0.4 g/kg body weight/day for 5 consecutive days). All but one patient with CIDP showed good functional improvement on IVIg during follow-up. The nonresponder received a treatment course of oral prednisone, 100 mg/day for 4 consecutive weeks. The patient improved with this therapy, and prednisone was tapered down over a 5-month period to 30 mg on alternate days.

The GBS patients did not show any deterioration during follow-up. After initial improvement, all 12 IVIg-responsive CIDP patients showed deterioration in clinical condition. Consequently, maintenance therapy with IVIg (1–2 days 0.4 g/kg/day; intervals, 3–21 weeks) was needed to prevent further deterioration and to regain earlier-achieved improvement. Eventually, this resulted in a general improvement of GS and reduction in arm disability during follow-up (Figs. 3A and 3B). A significant association was demonstrated between GS and arm-disability-scale values in these patients (Figs. 3C and 3D).

The calculated SRM scores for GS were good (SRM values for the right hands: 1.2, 1.1, 1.3, 1.2, and 0.8; SRM values for the left hands: 1.0, 0.8, 1.2, 1.3, and 1.1 at weeks 4, 12, 26, 40, and 52, respectively). The patients graded their clinical condition 51 times as “deteriorating,” 34 times as “stable,” and

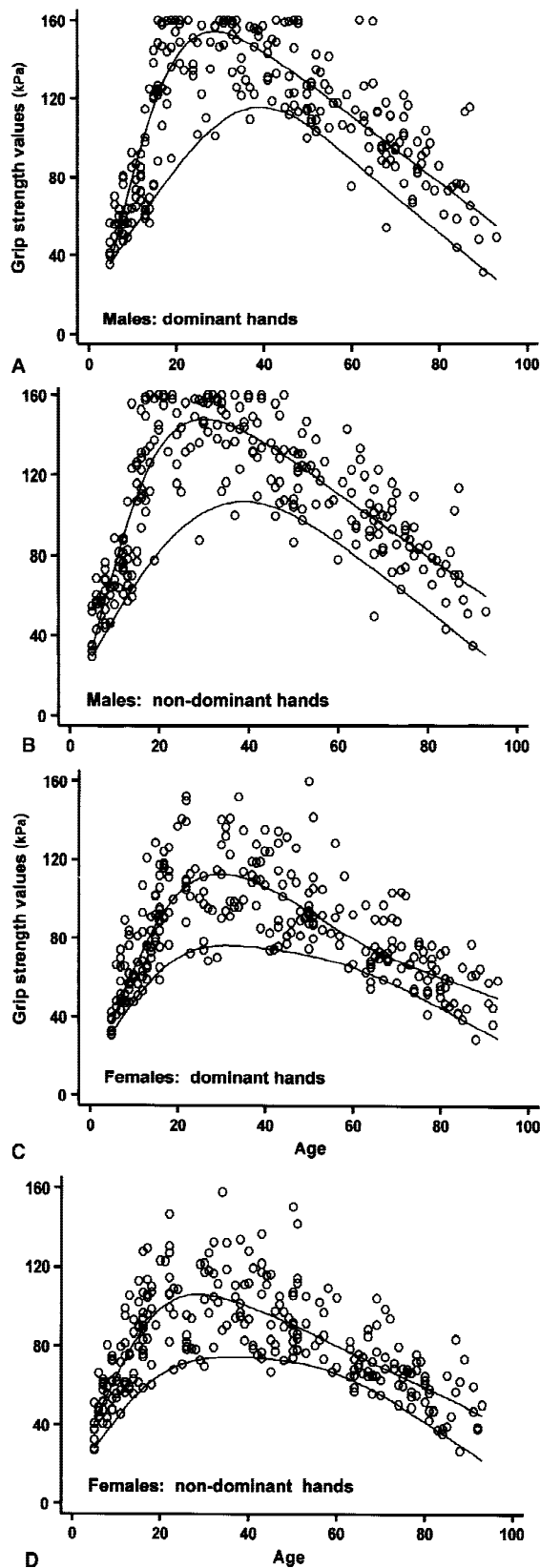


Table 2. Normal values for quantitative GS assessment with the Vigorimeter for both hands in healthy subjects (kPa).

Age (yr)	Males		Females	
	0.05 Quantile values per age span	Median values per age span	0.05 Quantile values per age span	Median values per age span
5-9	42	53	38	48
10-14	59	95	53	73
15-19	72	122	63	90
20-24	89	146	72	106
25-29	101	154	76	112
30-34	110	153	76	112
35-39	115	148	76	109
40-44	115	141	74	104
45-49	111	134	73	98
50-54	105	126	71	92
55-59	94	116	67	84
60-64	83	106	63	76
65-69	74	97	58	71
70-74	66	91	54	67
75-79	57	82	48	63
80-84	50	75	43	59
≥85	37	64	35	54

101 times as “improving.” These values were significantly associated with the serially obtained GS differences in these patients (random-effects linear regression analyses: $r = 0.52$ for association with right hands; $r = 0.47$ for association with left hands; $P \leq 0.003$).

DISCUSSION

The present study provides reference values for GS in males and females for both hands using the small bulb in healthy children up to 10 years of age and the medium-sized bulb of the hand-held Vigorimeter in healthy participants 10 years and older. In contrast to what is usually assumed in the literature, these reference values are not straight linearly or quadratic related to age and gender. The relation described in various papers is most probably due to the smaller age ranges studied, thus possibly presenting an incomplete view of GS in relation to age.^{8,9,12,14,24,30,34} Desrosiers and associates^{8,9} reported on two occasions normative data in healthy

FIGURE 2. Grip strength values with the Vigorimeter for the dominant and nondominant hands in healthy males ($n = 255$, 89% right-handed) and females ($n = 275$, 92% right-handed). The upper lines in each graph correspond to the calculated median GS values. The lower lines in each graph represent the 0.05 quantile reference values. These values were obtained in each gender separately using quantile regressions with restricted cubic spline functions.^{16,18}

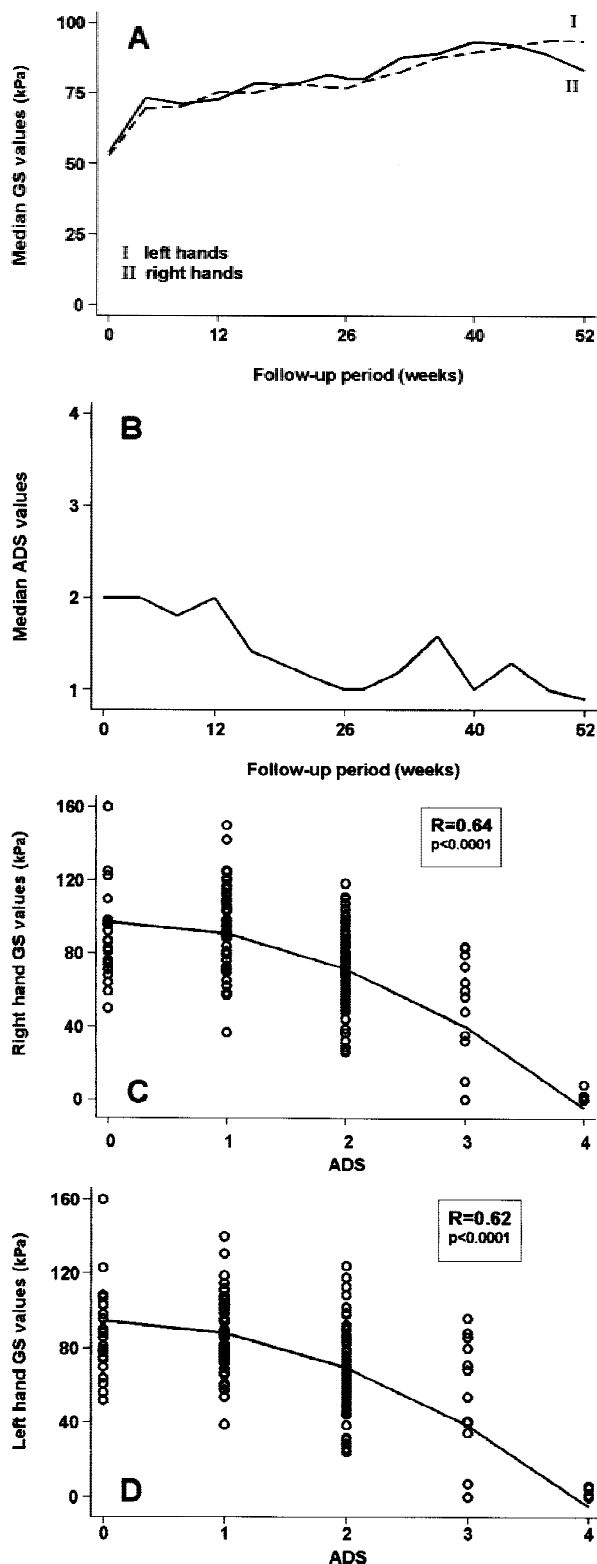


FIGURE 3. Changes of median GS for both hands (A) and median ADS scores (B) over time in 20 patients (7 GBS; 13 CIDP). The association between GS and ADS values are presented in (C) and (D) using "xtreg" (see section statistics).¹⁰ A reduction in ADS corresponds with improvement. One hundred eighty-six visits were completed.

individuals aged 60 years and older and suggested an association between quadratic age and GS. In another study, 486 healthy adults aged 16 to 94 years were examined, but unfortunately there was no stratification for age and gender and no information was provided regarding the healthy state of these adults.¹² To our knowledge, no study has provided reference values for the Vigorimeter in a cohort of healthy controls with an age ranging from 5 to 93 years. However, an equivalent study using the Jamar dynamometer was reported by Mathiowetz.²⁵ The graphical presentation of the reference GS values of this device demonstrated approximately the same shape as that seen in the current study for both genders.²⁵ Unfortunately, as in many studies, these values were obtained by calculating the means of sequential age portions in healthy participants.^{12,14,24,25,30,34} Thus, no consideration was given to the non-Gaussian distribution of the obtained GS values, as recommended by Altman² for these kind of data.

In addition to age, multivariate quantile-regression analyses demonstrated that hand circumference was a significant predictor of GS in males but not in females. The obtained GS values in the current study were consistently higher in men than in women. This is not surprising, because anthropometric variables were also constantly higher in men (Table 1).

The Vigorimeter was demonstrated in the current study to be easily applicable, quick, and user friendly, with significant correlation to a functional-arm scale. Good correlation was also demonstrated between this device and the widely used Jamar dynamometer.^{8,9,12} There are, however, mechanical differences between these two instruments that accounted for the preferential use of the Vigorimeter in the current study. We have previously applied the Jamar dynamometer in a group of patients with polyneuropathies and predominantly distal weakness. These patients consistently reported discomfort when assessing GS with this device. As in arthritic patients, the Jamar dynamometer caused stress on joints and skin and was less convenient because of its weight and rigidity.^{12,24} High reliability values were demonstrated for the Vigorimeter. The reproducibility of GS was independent of experience with the Vigorimeter, because the reliability values did not differ between examiners having different degrees of experience. Solgaard et al.³² investigated the accuracy of the Vigorimeter, using a "universal testing machine" that compressed the Vigorimeter with known forces, and also concluded that it was "very precise."

The usefulness of an outcome instrument depends not only on its simplicity, applicability, valid-

ity, and reliability but also on its ability to be responsive to clinical changes over time.^{13,22} A heuristic and statistic approach in examining responsiveness of a measure has been proposed.²² Statistical responsiveness captures the ability of an instrument to measure any change, irrespective of its relevance. Heuristic techniques are based on comparing changes as assessed by an outcome measure with an external indicator, for example, the clinical-judgment scores by the patient in the current study.^{13,22} We examined these two approaches in the responsiveness evaluation of GS. In the longitudinally examined patients, improvement of GS was significantly related to an external criterion, the patients' clinical-judgment scores, and the SRM scores for GS were equal to or higher than the proposed value of 0.8.^{7,23} However, these SRM values represent responsiveness within a group of patients. It is not clear whether substantial discriminative-responsiveness scores will be obtained for the Vigorimeter when evaluating various groups of patients, e.g., in a trial setting comparing a placebo versus a treated group.¹⁷

Grip strength was also significantly associated with the arm-disability values over time. These findings imply that, although a simple function in itself, GS can be applied as an index of arm-function recovery in patients with immune-mediated polyneuropathies. Hence, GS provides information on impairment and indirectly at disability level of outcome.

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